The Snow Energy Balance During Extreme Ablation: A Case Study of the March 2026 Heatwave in the Sierra Nevada

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Key Points:

- Record heat coincided with record rates of snow ablation during March of 2026, leading to record low-to-no-snowpack conditions.
- Net-shortwave was the dominant ablation driver despite the anomalously warm air temperatures.
- Low snow-albedo and higher than average downwelling shortwave due to a lack of mid level clouds contributed to record rates of net-shortwave of the snowpack.
- Snow ablated more slowly under vegetation canopy due to enhanced solar shading.
- Less than TODO: x of total snow ablation was due to sublimation.
- Boundary layer decoupling and vapor pressure arguments explain the small contribution of sensible heating and sublimation to the snow energy and mass balance during the heatwaves.

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Abstract

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1 Introduction

While meltwater from seasonal snowpacks has historically provided ∼70% of runoff in the western USA (Li et al., 2017), these natural reservoirs and our ability to measure, monitor, and predict them are in decline (Hale et al., 2023; Cowherd et al., 2024). From a practical perspective, many operational snowmelt runoff models use a temperature index approach, whereby air temperature (T_{air}) is the primary predictor of melt (Anderson, 1973). While these models have proven effective, the physical justification for such models is not clear. Since at least the 1980s the field of snow hydrology appreciates that understanding the climatic behavior of snow necessitates considering the entire energy budget of the snowpack. In particular, it is well accepted that the net shortwave radiation absorbed by the snowpack (SW_{net}) is a first-order predictor of rates of melt (Marks & Dozier, 1992; Cline, 1997). Both small impurities (Deems et al., 2013) and wet and dry grain-growth mechanisms (Colbeck, 1982) tend to decrease the amount of reflected sunlight (the snow albedo, α) and are responsible, in conjunction with rising sun-angles, for the large increases in SW_{net} in the spring. Unlike downwelling longwave ($LW\downarrow$; 4–100μm), the downwelling component of shortwave ($SW\downarrow$; 0.3–3μm) is mainly insensitive to variations in T_{air} itself. Variations in $SW\downarrow$ in time (and across elevation) are driven primarily by Rayleigh scattering by O_2 and N_2, Mie scattering by natural occurring and human-made aerosols, dust, and cloud droplets, as well as absorption primarily by water vapor, CO_2, and O_3.

At the same time, recent work has highlighted the strong correlation between extreme snow ablation (the loss of snow to either sublimation or melt) events and so-called "snow-eater heatwaves" — prolonged periods of sustained (day and night), anomalous, and above freezing T_{air} during the snow season (Rhoades et al., 2026). The characteristics of snow-eater heatwaves have already appreciably changed since the 1850s. These events are occurring more frequently, at a higher intensity, over a larger portion of the western US (WUS), and earlier in the water year.

An example of such an early-season, high-intensity, snow-eater heatwave (SEHW) occurred in March of 2026, when the monthly March T_{air} across the WUS were 3 standard deviations above long-term averages observed at many stations. Recent rapid-attribution studies have shown the snow loss associated with this event is TODO: X to Y % more likely due to anthropogenic climate change. This event (which started earlier than the window of occurrence studied in Rhoades et al. (2026)) resulted in widespread and rapid ablation at many snow-observing sites and one-to-two month earlier than average streamflow in many watersheds. This raises an apparent paradox: if $SW\downarrow$ provides most of the energy for snowmelt, and yet it is largely insensitive to T_{air}, then why are heatwaves so efficient at ablating snow?

Using the March 2026 event as a case study, we ask the following questions:

- What are the dominant snowmelt energy forcings during early-season heatwaves, and how do they scale with increasing T_{air}?
- How do vegetation canopies and atmospheric conditions — including clouds or lack thereof — shape such energetic forcings during heatwave events?

- What are the relative contributions of sublimation and melt to extreme snow ablation during such events?

For some perspective, approximately half of North America's snow water volume resides under tree canopy (Kim et al., 2021). Canopies absorb/scatter SW radiation, increase LW radiation, and reduce winds relative to open or non-vegetated snow surfaces; however, the magnitudes of these changes can vary widely across space and time. A number of studies have considered the effects of vegetation and forest treatments on the mean-state of the snow climate of this region, and it is well observed that snow tends to accumulate more and ablate later in the year in open regions relative to forested regions (Harpold et al., 2020; Krogh et al., 2020). Synthesis work has shown that open versus subcanopy snow dynamics are dependent on latitude and mean winter T_{air} (Lundquist et al., 2013). However, the interactions between vegetation and discrete extreme heatwave events has not previously been considered.

Despite otherwise drier-than-normal conditions, high-altitude cirrus clouds can occur during high-pressure ridging events commonly associated with cool-season heatwaves (Rhoades et al. (2026) Figure 1 and Supplemental Figure S5), however such events may preclude the formation of low or midlevel clouds, which have the largest impact on the snow energy balance (Rudisill et al., 2025) and are generally associated with reducing rates of snow ablation Frameworks for investigating interactions between clouds and surface radiation are well explored in the literature, using satellite retrievals and surface radiation data (Ramanathan et al., 1989; Shupe & Intrieri, 2004; L'Ecuyer et al., 2019), but little work has investigated the extent to which clouds or the lack-thereof contribute to anomalous snow ablation in mid-latitude regions.

The outcomes from this study answer basic science questions about energy available for snowmelt or sublimation in water-resource critical basins in the Sierra Nevada, with implications for other regions. The findings can directly inform efforts towards management and resilience against extreme events by providing actionable steps towards improving the monitoring and prediction of snowmelt. Last, improved understanding of vegetation and snow interactions during heatwaves can inform headwater management practices through improved decision support tools (e.g., Heggli et al. (2026)).

2 Background and Methods

This work is based around in-situ observations collected at the Central Sierra Snow Laboratory (CSSL; Schwartz et al. (2026)), described in the Sec. 2.1. Geographic controls on snow processes undoubtedly shape heatwave responses — but modeling these effects adds inherent uncertainties given the difficulty of distributing forcings across complex terrain TODO: CITE.

First, we place the record March 2026 heat and associated snow ablation in a climatic and synoptic context, using reanalysis data, remote sensing, and station observations (described in Sec. 2.2 and 2.3). To explain energy balance anomalies leading to the March ablation, we run single-point configurations of the Structure for Unifying Multiple Modeling Alternatives (Clark et al., 2015) from 2000-2026 (described in Sec. 2.4) for a forested and open site. Snow energy balance anomalies are interpreted in reference to historical March conditions for each site. Sec. 2.5——2.5.1 reviews the snow energy balance and effects of canopy, and Sec. 2.5.2 details the methods to quantify the contribution to ablation by radiative processes.

2.1 Study area

We focus our analysis on three basins in the immediate vicinity of the Central Sierra Snow Laboratory (CSSL; 39.32 °N, 120.37 °W, 2100 m.a.s.l) located in Soda Springs, CA (Schwartz et al., 2026). These include the Yuba, North Fork American, and Truckee River Watersheds (the greater Truckee River Watershed is composed of the Lake Tahoe basin and Truckee River basin at the eight-digit hydrologic unit code (HUC8) scale) (Fig. 1a–d). The Yuba and North Fork American rivers support the New Bullards Bar and Folsom reservoirs, respectively. These reservoirs have combined storage capacities approaching 2 million acre feet and hydropower generation potential of 559 megawatts TODO: citation. All three rivers support varied downstream consumptive uses (i.e., urban, industrial and agricultural) and provide critical aquatic and riparian habitats.

2.2 Observational data

We use data from twelve Natural Resources Conservation Service (NRCS) SNOwpack TELemetry (SNOTEL) Network stations located throughout the study region spanning TODO: X to TODO: Y m.a.s.l. SNOTEL is the primary snow observing program in the WUS (Serreze et al., 1999) and routinely measures snow water equivalent (SWE), precipitation, and T_{air}, though the homogeneity of T_{air} measurements remains a challenge for the SNOTEL network (Oyler et al., 2015; McAfee et al., 2019). To supplement SNOTEL temperature records, we use data from four US Historical Climate Reference Network (USHCRN) stations and the gridded nClimGrid product throughout the study region. USHCRN stations are high-quality stations subset from the larger GHCN-D dataset Menne et al. (2012). The HCNv2 dataset provides homogenized and quality-assured monthly T_{air} based on USHCRN data that dates as far back as the late 1800s (Menne et al., 2009). While this data does not have the same spatial density and elevation range as SNOTEL, the longer temporal record and high quality control mean that these observations can more confidently be used for evaluation of climate extremes, such as heatwave events (Bercos-Hickey et al., 2022).

In addition to hosting an NRCS SNOTEL site, CSSL provides long-term meteorological and snowpack records in addition to surface energy balance measurements. Campbell Scientific CNR4 four-component radiometers measure incoming and outgoing SW and LW radiation from the atmosphere and snowpack. A Campbell Scientific IRGASON eddy covariance system measures sensible (H_s) and latent (H_l) heat fluxes above the snowpack. The CSSL forest study site hosts an additional meteorological station approximately 150 m north of the main CSSL site. This tower hosts an additional Kipp and Zonen SP Lite2 radiometer that measures below-canopy up and downwelling shortwave radiation. SWE measurements were obtained using a federal sampler (Osterhuber, 2014) at nine discrete sampling intervals during the March heatwave in the forest location. To maintain temporally and serially complete records and to compare against climatological March conditions at this site, this work relies primarily on energy balance model estimates from SUMMA, which we validate in detail against all-available CSSL energy balance fluxes when available (Sec. 2.4.2).

2.3 Atmospheric structure and radiation data

To investigate the synoptic patterns associated with the March heatwave, we use 500 hPa geopotential height and atmospheric column information from the fifth-generation ECMWF Reanalysis (ERA5) (Hersbach et al., 2020). Daily geopotential height anomalies during the event are computed using data provided by Lee et al. (2023), which has been de-trended and covers the period from 1979 to 2022.

We use the Clouds and the Earth's Radiant Energy System (CERES) EBAF, hereafter CERES-EBAF (Kato et al., 2018) to investigate the effects of clouds or lack-thereof on the snow energy balance. CERES-EBAF provides gold-standard measurements of surface and top-of-atmosphere (TOA) irradiances at a 1x1 degree resolution at a monthly cadence. This product is suitable for trend analysis. Rudisill et al. (2025) compared cloud radiative effects from CERES products against in-situ radiometers collected by the SAIL field campaign in the Colorado Rockies (Feldman et al., 2023) and found good agreement, despite grid-resolution mismatches and confounding effects of terrain. Hinkelman et al. (2015) likewise found the lower level, hourly CERES-SYN1Deg (Rutan et al., 2015) product proved efficacious when used as forcings for snow simulations. The 2000–present period-of-record of CERES-EBAF is used to examine the March anomaly in $SW\downarrow$ and $LW\downarrow$ and the contribution of clouds to the anomalies therein.

2.4 SUMMA model configuration

To investigate the snow energy balance, we use the SUMMA model (Clark et al., 2015). SUMMA is well-vetted for a range of snow climate applications and provides several options for flux stability corrections, canopy radiation interactions, snow-layering schemes, and other model structural options (Cristea et al., 2022). We use the Jordan snow layering option, which allows for up to 100 snow layers, the UEB-2 stream approximation for canopy radiation interactions, the "Louisinv" option for stability correction of turbulent fluxes, and the "stickySnow" canopy snow interception scheme. SUMMA solves for a vegetation energy balance and canopy air-space temperature which is used to produce $LW\downarrow$ incident onto the sub-canopy snowpack. Additional specific modeling parameterizations with relevance to this study are described in Sec. 2.5 and the Supplemental Material. We assign a measurement height of 12m for the model forcings into SUMMA in order to intercompare the effects of 10m tall needleleaf vegetation characteristic of the forest site at CSSL using the same forcing variables.

We use hourly ERA5 data (Hersbach et al., 2020) as SUMMA model forcings after adjusting to the CSSL site. Daily precipitation from the NRCS SNOTEL site is used directly as forcing when available. Hourly ERA5 two-meter T_{air} are bias corrected to match the monthly average T_{air} observed at the CSSL from the NRCS records by removing a constant linear bias term. Comparing ERA5 $SW\downarrow$ data to observations from the CNR4 shows good agreement except for morning and evening when the surrounding vegetation blocks direct beam solar radiation. To account for this effect, we compute and apply a $SW\downarrow$ correction coefficient as a function of the solar zenith and azimuth angles to adjust the ERA5 data to account for shadows (see Figure S3). To ensure that the ERA5 data matches the best available estimates of variability and trends in radiation at this site, we use a multiplicative scale that forces the ERA5 $SW\downarrow$ and $LW\downarrow$ monthly anomalies to match monthly anomalies in the CERES-EBAF data. Other forcing variables compared favorably against the available observations and were not adjusted. Additional information about SUMMA validation is provided in the Supplemental Material (Sec. 2.4.2).

2.5 The snow energy balance

Using meteorological sign conventions for turbulent fluxes, the energy budget of a snowpack is given by:

$$\mathrm{Melt/Refreeze} - \frac{d}{dt}\mathrm{CC} = \underbrace{H_{\mathrm{s}} + H_{\mathrm{l}}}_{\text{turbulent fluxes}} + \underbrace{LW\downarrow - \sigma\epsilon T_{snow}^4}_{\text{net longwave}} + \underbrace{SW\downarrow(1-\alpha)}_{\text{net shortwave}} + G \qquad (1)$$

,

where Melt/Refreeze is the rate of melt or liquid water refreezing in mass per second multiplied by the latent heat of fusion, $\frac{d}{dt}\mathrm{CC}$ is the time rate-of-change in the snowpack cold-content (Jennings et al., 2018), H_s and H_l are the turbulent fluxes of sensible and latent heat respectively at the snow/atmosphere interface, T_{air} is the temperature of the air, T_{snow} is the temperature of the snow surface, ϵ is the emissivity of the snow (0.98), $SW\downarrow$ is the downwelling shortwave radiation, and α is the albedo of the snow surface defined as the ratio of upwelling shortwave radiation ($SW\uparrow$) to $SW\downarrow$.

The mass-balance of the snowpack is related to the energy balance (Eq. 1) in the following ways. The daily or monthly change in SWE (referred to as ΔSWE) is related to both Melt and to a lesser extent H_l, though the latter may remove 10% to upwards of 28% of the seasonal snow mass depending on a variety of factors (Sexstone et al., 2018; Schwat et al., 2025). Melt may be either positive (melt) or negative (refreezing) using the sign convention in Eq. 1. Melt integrated throughout an entire day is not synonymous with a loss in SWE, since meltwater may be retained in the snowpack and may refreeze or remain in place by tension. For flow through the bottom boundary of the snowpack, the dry pore space of the snow must be filled with liquid water to residual saturation (Colbeck, 1976; Seligman et al., 2014). Snowpacks may retain between 2–8% liquid water by volume. In the SUMMA model, meltwater flow throughout the snowpack layers is modeled as a gravity-driven power law; preferential flow is not considered.

2.5.1 The influence of canopy

Vegetation canopy influences the snow energy balance through several mechanisms. First, falling snow is intercepted by the canopy and may melt or sublimate from the canopy before reaching the snowpack, so total SWE accumulation under canopy is commonly less than adjacent open areas (Musselman et al., 2008). Second, canopy layers may inhibit turbulent exchange of mass and energy by suppressing wind. Third, vegetation impacts the radiative balance of the snowpack by altering $SW\downarrow$ and $LW\downarrow$ impacting the snow surface. In SUMMA, the incident $LW\downarrow$ to the sub-canopy snowpack is modelled by:

$$LW\downarrow_{\text{under canopy}} = \underbrace{(1-\varepsilon_c)\,LW\downarrow_{\text{atm}}}_{\text{sky LW through canopy gaps}} + \underbrace{\varepsilon_c\,\sigma\,T_c^4}_{\text{canopy emission}} \tag{2}$$

, where ϵ_c is the emissivity of the canopy layer. In SUMMA, the ϵ_c is purely a function of the leaf-area index (LAI) and stem-area index. We turn off vegetation phenology in the model such that the ϵ_c is constant for these experiments. For $SW\downarrow$ below the canopy, we use the "UEB-2stream" SUMMA parameterization which is based on Mahat and Tarboton (2012). An LAI value of 1.5 was assigned based on data from nearby representative regions (Krogh et al., 2020). These choices are further validated by comparing modelled sub-canopy $SW\downarrow$ for a range of LAI values against the CSSL radiometers in the forest site, and finding good agreement, supporting these parameterization choices (Fig. S5).

2.5.2 Defining radiative forcing and effect

We apply the radiative forcing concept in several ways in this study (Ramanathan et al., 1989; Deems et al., 2013; Rudisill et al., 2025). In this context, radiative forcing quantifies the instantaneous difference between the net or a single stream of radiation between observed conditions and a counterfactual reference state.

We define RF_{canopy} as the difference between R_{net} under the canopy relative to R_{net} outside of the canopy layer. Lacking observations sufficient to constrain RF_{canopy}, we use the SUMMA model to evaluate this quantity and express this value for both LW and SW components. To investigate the relative contribution of α decay on the surface radiation balance, we quantify the α radiative forcing (RF_{albedo}) by differencing SW_{net} computed using observed α versus SW_{net} computed using observed $SW\downarrow$ and a hypothetical fresh-snow broadband α; for this work we use 0.85.

We define a cloud radiative "effect", rather than "forcing" to express the influence of clouds on downwelling $LW\downarrow$ and $SW\downarrow$ terms only (RE_{cloud}), since a full accounting of cloud radiative effects on net-energy terms also depends on the amount reflected and absorbed by the surface, and therefore the α (Rudisill et al., 2025) and assumptions about varying upwelling emissions. All radiative effects are signed such that positive indicates energy is added to the snowpack via that process (clouds, canopy, or α decay), and a negative sign indicates the opposite.

Table 1. Definitions of radiative effects used in this study. All effects are signed positive when energy is added to the snowpack. x refers to the radiative component, which is either SW, LW, or net (the sum of SW and LW).

Name	Abbreviation	Equation
Cloud Radiative Effect	RE_{cloud}	$F_{x,cloudy}\downarrow$ - $F_{x,clear}\downarrow$
Canopy Radiative Forcing	RF_{canopy}	$F_{x,\ under\text{-}canopy}$ - $F_{x,\ open}$
Albedo Radiative Forcing	RF_{albedo}	$SW\downarrow[(1\text{-}\alpha_{obs})$ - $(1\text{-}\alpha_{clean})]$

2.6 Ranking SWE and heatwave anomalies

To rank or contextualize SWE anomalies, we use the framework from Hatchett et al. (2022), which uses a percentile-based metric to assign snow drought categories. Drought categories are ranked as abnormally dry (D0), moderate drought (D1), severe drought (D2), extreme drought (D3), and exceptional drought (D4) for values between the 30th–20th, 20th–10th, 10th–5th, and 5th–2nd and below the 2nd percentiles, respectively. SWE percentiles are computed using the NRCS SNOTEL periods of record for each site.

Z-scores are used to rank near-surface T_{air} and snow ablation anomalies, given by

$$z = \frac{X-\mu}{\sigma}$$

. Baseline mean and standard deviations are computed using a common 1981–2020 normal period unless otherwise stated.

3 Results

3.1 The March 2026 SEHW

A four-station composite of the long-term and homogenized monthly USHCRN HCNv2-data shows that 2026 had the highest March T_{air} observed since the beginning of the record in 1890, and was +3.6 °C above the 1981–2020 average — a full three-σ above the mean (z=3) (Fig. 2a). The nClimGrid likewise shows a +4–6 °C anomaly above average in some locations, compared to the +3.6°C anomaly observed at the four-station USHCRN composite (Fig. 2b).

This prolonged period of heat resulted from an extreme and persistent ridge of high pressure that gained strength throughout March (Fig. 3a,b), with 500 hPa geopotential height anomalies exceeding two standard deviations and 200m throughout March.

Immediately prior to the March 2026 heatwave, the region experienced three notable periods of weather. While the water-year-to-date precipitation (1 October–28 February) was 100% of 1991–2020 normal leading up to March 2026 (not shown), there was a prolonged midwinter dry spell (Hatchett et al., 2023) from 7 January to 9 February during which SWE began ablating (Fig. 3c). Then, between February 14th to 19th, a total of 100 mm of new SWE accumulated at CSSL, the third-highest four-day total since records began in 1971. This snowfall event was followed by anomalously warm T_{air} and a rain-on-snow (ROS) event on the 24–25 of February that brought 91 mm of rainfall, leading to a net decline in SWE (Fig. 3b,c).

The beginning of March started with average to slightly above average T_{air}. As the high pressure ridge grew over the WUS, daily maximum T_{air} peaked at $22.2°C$ on 19 March and $23.3°C$ on 20 March at CSSL TODO: report daily avg air temp but remained well-above normal from 8–31 March, with daily T_{air} values at the Tahoe City COOP station yielding anomalies of z=2 or greater for four days between the 17th–27th. Correspondingly, the 500 hPa height anomaly exceeded 200 meters across the WUS on this date, and was greater than two-σ above the detrended mean for this date. The hemispheric wave pattern on 16 March in particular is characteristic of an amplified North American Dipole pattern (S.-Y. Wang et al., 2014; S.-Y. S. Wang et al., 2015; Singh et al., 2016).

Snow began ablating consistently starting with the February ROS event for both the open and forest sites and continues into the month of March. We find good agreement between both SUMMA and observed SWE (Fig. 3c) and rates of ablation. SWE ablated at an average rate of 7.37 mm/day (7.06 mm/day) between the first of March and the 14th of March, after which it accelerated to a rate of 25.30 mm/day (18.66 mm/day) until the end of the month in the open (forest) until ablating completely in the open by the TODO: 28th and the forest site by the TODO: 24th. Modeling results also show that, starting on March 1, the snow contained significant liquid water throughout the depth of the pack on the order of 1-2% by volume due to the antecedent ROS event. A small snow event occured shortly after on the

3.2 SWE anomalies during water year 2026

SWE completely ablated earlier, and at a faster rate than typical for the time period, at northern Sierra SNOTELs (Fig. 4a–c). The beginning of March saw SWE levels at a D1 or D0 ("no drought") level across the twelve SNOTEL sites (Fig. 4d). While SWE was nearly 0 mm for two of the lower-elevation sites (station ID numbers 473 and 809) this was not particularly anomalous for those locations for this date. By the end of March, 9 of the 12 stations reached 0 mm SWE and the remaining sites were in a D3 ("extreme drought") drought classification (Fig. 4e). Only 4 of the SNOTEL sites had a precedent for 0 mm of SWE on 31 March. Similarly, the snow covered area reached its lowest ever recorded value on 31 March during the Moderate Resolution Imaging Spectroradiometer (MODIS) record (see Supplemental Material; Fig. **??**). These findings are consistent with the observation that 17% of SNOTELs had the lowest 15 March, 2026 SWE on record (Marshall et al., 2026).

An important distinction, however, is that rate of monthly snow ablation was also anomalous at most sites (Fig. 4f). Typically, most SNOTEL sites in this region gain (or remain constant) rather than lose SWE throughout March (Fig. 4b) with the exception of two of the lowest elevation sites. Nine out of 12 sites, including CSSL, measured their highest rate of monthly SWE ablation for the SNOTEL record. The lowest elevation site (473) is the only site without anomalous monthly SWE ablation, in part because the starting value was so close to zero.

3.3 The snow energy balance in the open and forest site

To understand the causes of the anomalous observed ablation described in Sec. 3.2, we now focus on the CSSL site (site ID 428 in Fig. 4 and also depicted in Fig. 3c). These causes, in addition to the different rates of ablation observed at the forest and open sites, can be understood by examining a time series of energy balance components (Eq. 1) and by separating them into nighttime, daytime, and 24-hour averages (Fig. 5a–f, Fig. 6a–f).

H_s and H_l are relatively minor factors compared to radiative fluxes (Fig. 5a,b, Fig. 6a,b). Two periods of significant sublimation occurred (negative H_l) on the 5th to the 7th, which correspond with relatively high wind speed (Fig. 3c) but prior to the onset of the warmest March temperatures. Another significant sublimation event took place on the 24th. The highest rates of H_s occurred between the 3rd through the 10th and again from the 21st to the 26th. A maximum daytime H_s exceeding 80 $W{\cdot}m^{-2}$ took place on the 22nd for both the forest and open sites, just one day after the peak of the March temperatures.

SW_{net} far exceeds the H_s maxima in the open site, however (Fig. 5c, Fig. 6c). Here, SW_{net} increases from daytime values approaching 80 $\mathrm{W{\cdot}m^{-2}}$ to values exceeding 240 $\mathrm{W{\cdot}m^{-2}}$ by the end of the month (with 24hr averages of approximately 120 $\mathrm{W{\cdot}m^{-2}}$). Rates are significantly lower in the forest. SW_{net} increases throughout March, but never exceeds 80 $\mathrm{W{\cdot}m^{-2}}$.

In the forest, LW_{net} is positive on average for the entirety of March 2026 except the 4th–7th, which also corresponds with the only period of significant overnight cold-content production (Fig. 5d, Fig. 6d). Overnight LW_{net} switches from slight negative ($\approx$ -8 $\mathrm{W{\cdot}m^{-2}}$) to positive or near-zero by the 16th in the forest site. The highest rate of LW_{net} occurs on the 22nd, two days after the peak of the heatwave event, and exceeds 60 $\mathrm{W{\cdot}m^{-2}}$ during the daytime.

LW_{net} in the open site has a completely different sign, meaning that the snow is continuously losing heat via longwave emission, even during the peak of the heatwave in the latter half of March, and on the order of -80 to -40 $\mathrm{W{\cdot}m^{-2}}$.

The previous energy exchanges, in addition to ground heat flux (not shown), cumulatively manifest as energy directed towards breaking/reforming ice crystalline bonds (Melt/Refreeze) and warming ice to the melting point (ΔCC). A significant amount of energy is re-added back to the snowpack during the night by liquid water refreezing (6–33 $\mathrm{W{\cdot}m^{-2}}$) in the open site, and to a lesser extent the forest site (Fig. 5e, Fig. 6e). The overnight LW_{net} cooling in the open yields overnight ΔCC that is eroded the following day, prior to the initiation of melt (yielding net-zero averages over a 24 hour period; Fig. 5f). The cumulative overnight production in cold-content becomes small compared to the following day's melt by the 14th of March and represents less than 20% of the total melt energy, indicating that the lack of overnight cold-content production — almost purely driven by the decline in LW_{net} and increase in overnight H_s— yields a snowpack that is primed for ablation the following day in the open site.

The dynamics of ΔCC are different in the forest — there is a near-zero amount of overnight cold-content production and very little refreezing. These model results are confirmed by field reports from CSSL, who found melt-refreeze crusts in the open during March but slush in the forest site even during morning observations, in agreement with these results.

3.4 Radiative forcing by canopy and snow ageing

The radiative forcing concept (Sec. 2.5.2) can be used to further interpret daily and monthly variation in the energy fluxes described previously and shed light on the differences between the open and forest site (Fig. 7a–f).

Both RF_{albedo} and RF_{canopy} have a demonstrable impact on the energy balance. Snow α started at a relatively low value leading up to the March heatwave (0.6 to 0.7) and declines consistently during the month of March (Fig. 7a). The declines are well-matched by remote sensing and CSSL observations (see Supplemental Material).

RF_{albedo} increases at the open site as a function of both rising sun-angle and $SW\downarrow$ (not shown) and α decline, reaching a value exceeding 60 $\mathrm{W{\cdot}m^{-2}}$ in the open site based on the SUMMA simulation. While there is only a minor difference between the SUMMA model snow α between the forest and open sites, the RF_{albedo} in the open exceeds RF_{albedo} in the forest site by more than a factor of two by the end of the heatwave due to attenuation of $SW\downarrow$ by the canopy layer (Fig. 7b).

Second in terms of magnitude to RF_{albedo}, RF_{canopy} yields a +20 to -90 $\mathrm{W{\cdot}m^{-2}}$ maximum daily rate during the month (Fig. 7c). There are two competing effects. $\mathrm{RF}_{canopy\,,LW}$ increases as a function of T_{air} at a rate of 3.3 $\mathrm{W{\cdot}m^{-2}}$ per °C of T_{air}, which is visualized by considering the daily relationship for the entire SUMMA output record for the month of March (Fig. 7e). Additional analysis (not shown) indicates that the canopy layer is consistently warmer than the ambient T_{air} during the daytime, which combined with the emissivity relationship (Eq. 2) explains this effect. This effect does not exceed the shading effect of the canopy ($\mathrm{RF}_{canopy\ ,SW}$) however, even though the canopy attenuates much of the incident sunlight, so $\mathrm{RF}_{canopy\ ,Net}$ is near-zero and transitions to negative (cooling the snow) during the hottest part of the heatwave when the sun is also the most intense and the α is also lowest.

Ultimately, the cumulative energy transmitted from the atmosphere to the snow attributed to RF_{albedo} is +101.3 and +23.7 $\mathrm{MJ{\cdot}m^{-2}}$ for the open and forest site respectively and RF_{canopy} is responsible for a -95.1 $\mathrm{MJ{\cdot}m^{-2}}$ effect (reducing energy incident on the snowpack). We do not compute a similar value for the forest site since the co-variance with the vegetation energy and radiative transfer does not yield a pragmatic solution, but we hypothesize that the magnitude may be similar due to the competing effects from SW and LW mechanisms. RE_{cloud} is evaluated in Sec. 3.5.

3.5 Climatological context of the March 2026 snow energy balance

How does the March 2026 energy balance compare to typical Marches at CSSL? We have seen previously that, on average, SWE gains rather than ablates (Fig. 4f) during the month of March.

The daily components of the energy budget can be contextualized by examining cumulative sums, in $\mathrm{MJ{\cdot}m^{-2}}$, of the different energy balance terms to quantify their total contributions to ablation (Fig. 8a–g). For some perspective, 1 $\mathrm{MJ{\cdot}m^{-2}}$ of energy is equivalent to approximately 3.0 mm of equivalent snowmelt or 0.35 mm of snow sublimation (found by dividing by the latent heat of fusion and sublimation, respectively).

SW_{net} is both the strongest forcing (+223.6 $\mathrm{MJ{\cdot}m^{-2}}$) and also the most anomalous forcing compared to the 1980–present March average (+72.0 $\mathrm{MJ{\cdot}m^{-2}}$), followed by H_s (+45.6 $\mathrm{MJ{\cdot}m^{-2}}$, with a +18.7 anomaly) in the open site (Fig. 8a, Table 2). For the forest site, SW_{net} is also the strongest forcing (+48.3 $\mathrm{MJ{\cdot}m^{-2}}$) but not the most anomalous. The most anomalous forcing in the forest is LW_{net}, which is the second strongest forcing (+27.8 $\mathrm{MJ{\cdot}m^{-2}}$) with an anomaly of +28.9 $\mathrm{MJ{\cdot}m^{-2}}$ relative to historical conditions, indicating that — for a typical year — LW_{net} is below-zero (and a source of snow cooling) rather than providing energy for heating snow under canopy. The lack of a significant LW_{net} anomaly in the open relative to the anomalous $LW\downarrow$ is in part explained by the compensating effects of outgoing $LW\uparrow$— warmer snow radiates more energy exponentially (1).

The cumulative terms of panels Fig. 8a,d sum to the total Melt, with the small exception of the change in the initial snow cold-content (which is by definition zero when there is no snow) and is not depicted in the figure. The initial cold-content is small compared to the accumulated energy during the snow ablation period, and is on the order of 1 $\mathrm{MJ{\cdot}m^{-2}}$ for the open site. The mass loss associated with Melt, combined with mass loss via sublimation, is equivalent to the total Δ SWE during the event. Only 3.1 cumulatively sublimated from the forest site compared with 0.4 mm from the open site — which are less than 1% of the total ablation. In both sites this is less than half of the model estimated sublimation for typical March conditions (Table 2).

The above-canopy $SW\ \downarrow$ and $LW\ \downarrow$ (which are used as the top-boundary forcing for both open and canopy simulations) are also particularly anomalous during March 2026 (Fig. 8a). To understand these anomalies, we examine map-views and timeseries of the CERES-EBAF product for the WUS and northern Sierra Nevada subset region (Fig. 10a–c). March of 2026 had the highest $SW\downarrow$ to the entire CERES record dating back to March of 2000 and is more than 1.5 standard deviations above average for the subset region. Nearly 40 $\mathrm{W{\cdot}m^{-2}}$ of additional $SW\downarrow$ energy irradiated the land surface compared to average conditions. While less anomalous in terms of z-score, the $LW\downarrow$ anomaly is over 9 W·m^{-2} above average and the second-highest rate of monthly $LW\downarrow$ during the CERES-EBAF record for this region.

The variations in the $SW\downarrow$ anomaly are almost entirely driven by variations in cloud radiative effect (which itself depends on cloud amount and optical properties). The same is not true for $LW\downarrow$, which also depends on other factors including water vapor and column T_{air} (Fig. 10c,d). RE$_{cloud}$ values were the highest (i.e., the weakest rate of cooling) recorded by CERES-EBAF, and correspondingly the RE$_{cloud}$ LW effect was also the lowest on record, which together indicate that skies were anomalously free of optically thick clouds during this period. In other words, clouds typically provide about a 60 W·m^{-2} reduction in $SW\downarrow$ relative to clear skies in March, but only provided 20 W·m^{-2} during March. Correspondingly, no precipitation was observed during the month (Fig. 3c). With this in mind, it is even more remarkable that $LW\downarrow$ was above average for this region despite a clear-atmosphere, since clouds radiative as a nearly perfect gray-body at their base-temperature (Shupe & Intrieri, 2004).

In terms of energy, the anomalous increase $SW\downarrow$ due to the lack of optically thick clouds is equivalent to 93 MJ·m^{-2} integrated over the duration of snow cover in the open during March of 2026. Assuming 60% of that energy is reflected, this is equivalent to 37 MJ·m^{-2}, or 110 mm of melt assuming that all of this energy is converted to melt (as opposed to sublimation or other terms of the energy balance). The monthly cadence of the CERES-EBAF product does not allow us to discriminate if the peak of the anomaly corresponded with the early, cooler part of the month or the SEHW event later in the month.

4 Discussion

4.1 Sensible heat and sublimation do not scale linearly with $\mathbf{T}_{air}$

The relatively small contributions of H$_l$ and H$_s$ despite anomalously high T_{air} can be explained by considering "offline" test cases assuming a fixed 0 °C melting snow surface. Snow surface temperatures can never exceed this fundamental thermodynamic limit, leading to decoupling of the surface layer above the snow from the boundary layer air (Mott et al., 2013).

If a neutral condition is assumed without the effects of buoyant suppression, H$_s$ over melting 0 °C snow increases linearly as a function of T_{air} with a slope determined by w_{spd} (Fig. 11a). Accounting for buoyant suppression (using the same parameterization implemented in SUMMA; see TODO: supp) decreases H$_s$ relative to the neutral assumption, and so much so that a moderate constant w_{spd} value of 5 m·s^{-1} yields H$_s$ that actually *decreases* as a function of increasing T_{air} above approximately 4 °C.

Fig. 11c depicts the T_{air} value that maximized H$_s$ for a given w_{spd}. A w_{spd} of 2 m·s^{-1} achieves a maximum H$_s$ when T_{air} is only a degree or two above freezing. A T_{air} of 25 °C— slightly less than the daily maximum recorded at CSSL — requires a wind speed of over 10 m·s^{-1} to maximize H$_s$.

Evaporation and/or sublimation is likewise impacted by snow's 0 °C thermodynamic limit — the vapor pressure of pooling 0°C liquid water or ice on top of the snowpack never exceeds 6.11 hPa to a reasonable approximation. Therefore, the vapor pressure of the air must be less than 6.11 hPa for sublimation or evaporation to occur. For a constant relative humidity (RH) of 20% (roughly the minimum observed during the March heatwave), H$_l$ decreases rather than increases as a function of T_{air} over a 0°C snow regardless of w_{spd} (Fig. 11b,d). T_{air} greater than 24–25 °C at 20% RH *adds energy and mass to the snow* via condensation/deposition. On the other hand, sublimation is permitted across the full range of RH values as T_{air} approaches 0°C. In reality, though, RH is not independent of T_{air}, and generally decreases as T_{air} rises during the daytime due to water vapor conservation and dry-air entrainment from the upper levels of the boundary layer. RH values did reach these low levels for brief periods near noon, but were not sustained for long durations, explaining the relative lack of sublimation (less than TODO: 1% of the total snow mass loss during March). The effects of thermodynamic stability further reduce H_l as a function of T_{air} as well.

These arguments hold across elevation gradients with two additional considerations. All else being equal, H_s will decrease with elevation as the density of dry air decreases (ρ in Supplemental Eq. 1). On the other hand, conditions that favor sublimation may become more prevalent at higher elevations given the lapse rate of specific humidity and for snowpacks extending above the boundary layer.

Some warming experiments have shown increases in sublimation during mid-winter warm spells (Scaff et al., 2024), though others have found modest or no-change (Sexstone et al., 2018). Results from downscaled GCM have pointed towards increases in nocturnal sublimation in contintental snowpacks (Rudisill et al., 2026). Observations also show higher sublimation rates occurring later in the snow season (Schwat et al., 2025). This is still consistent with the previous arguments, however, which are made for the specific case of increasing heat over 0°C snow. Increased sublimation with warming may be more indicative of the snow itself warming and increasing it's vapor pressure (up to the 6.11 hPa limit at 0°C). In other words, snow that climatologically shifts in surface temperature closer to a 0°C state may more frequently sublimate into the atmosphere, but discrete heatwave events may cause a decrease in sublimation if the snow near it's vapor pressure limit.

The previous argument also assumes that Monin-Obukhov theory — which underpins turbulent exchange in nearly all land-models and climate models — is applicable. Mesoscale, terrain related flow features that generate turbulence from above the ground surface and provide additional mechanical turbulent mixing may invalidate the assumptions of this theory (Schwat et al., 2025). Likewise, the advection of sensible heat, such as from the forest edges and snow-free patches, may provide additional sensible/latent heat that is not accounted for (Harder et al., 2017). More work investigating turbulent exchange characteristics over snow during extreme heatwave conditions merits additional investigation.

4.2 Ablation and the relationship to overnight above freezing T_{air}

Whether the overnight T_{air} is above or below freezing is often considered an important factor explaining snow ablation, since refreezing overnight requires overcoming the energy required to re-break the ice-bonds the following day in order to generate liquid meltwater that may percolate out of the bottom of the snowpack. The snow does not instantly freeze when the two-meter T_{air} cools to 0 °C, but rather depends on the energy balance of the snow which sheds heat via LW_{net} and H_s (if T_{air} is warmer than the snow layer).

We found that overnight T_{air} is correlated with the energy balance in many ways, but it is not alone a sufficient metric to inform rates of snow ablation or the snow energy balance writ-large during the March 2026 ablation period (Fig. 12a,b). Overnight refreezing is actually better correlated with overnight $LW\downarrow$ than T_{air} alone (R^2 of 0.13 vs. 0.41), even across the above/below freezing T_{air} threshold, and the same is true for overnight cold-content production (R^2 of 0.06 vs. 0.44). Overnight average T_{air} and $LW\downarrow$ similarly explain variance in next-day rates of melt. We caution that this analysis is not intended as a predictive model, keeping in mind these are all highly correlated variables. But this finding does highlight the importance of $LW\downarrow$ measurements and model products, since $LW\downarrow$ is better correlated with the overnight energy balance of the snow than T_{air} alone.

4.3 Limitations and caveats related to the influence of canopy

We only examined a limited case of vegetation canopy with a fixed LAI of 1.5, which corresponded well with CSSL observations (Fig. S5) and is representative of other forested regions of the Sierra Nevada (Van Gunst, 2012; Krogh et al., 2020). A full accounting of the geographic pattern of vegetation/heatwave interactions likely requires considering additional processes such as canopy gap effects, shading from adjacent stands, and the full range of canopy types and parameters that are also aspect and slope dependent (Broxton et al., 2015).

Lundquist et al. (2013) presented a meta-analysis of snow climate studies and concluded that regions where climatological T_{air} exceed -1 °C typically observe earlier melt out in forested areas, relative to open regions, and attribute the cause to the interactions related to the concepts expressed by our RF_{canopy} analysis. They likewise conclude that forest snow is more sensitive to climate change in the Sierra Nevada than non-forested snow, given the temperature dependency of sub-canopy $LW \downarrow$ (Fig. 7e).

We showed that, while snow did melt out earlier under canopy compared to the open, it did so at a lower rate, including during the period of the warmest SEHW conditions. The cause was the antecedent snow-interception rather than an energy contribution, however, given the lower ablation rate. We have one important addition to Lundquist et al. (2013) concerning the climate sensitivity of snow under/out of vegetation canopy — for this extreme event, T_{air} was anomalously high in addition to $SW \downarrow$ due to the lack of clouds, on the order of +40 $\text{W}\cdot\text{m}^{-2}$, meaning that shading effects by the canopy were also enhanced (Fig. 10). Future monthly timescale $SW \downarrow$ anomalies are expected in the Sierra Nevada with warming. Some observational evidence is already pointing towards enhanced $SW \downarrow$ in continental regions writ-large, including snow climates (Nyeki et al., 2019).

While we can conclude that forest provided resiliency to the SEHW event in terms of reducing the rate-of-ablation, more work is required to investigate the covariance of T_{air} and $SW \downarrow$ anomalies during SEHW events; warm but cloudy conditions may yield the opposite trends in ablation rate under forest versus open. The relationships are also likely elevation and aspect depedent. In addition, a key limitation of this work is the parameterization of the effects of leaf-litter on the α of the snow under canopy, which is not parameterized in SUMMA. More work is required to understand these interactions to establish optimal forest configurations for snow-resiliency against extreme heat conditions.

4.4 Sensitivity to antecedent conditions

The timescales of anomalous ablation imply that different mechanisms are likely important. For instance, cold-content of the snow is vital for understanding event-scale ablation rates during ROS (Katz et al., 2023). While the February wet-SEHW and rain-on-snow event did lead to below average initial cold-content, reduce surface α, and saturate the snow with liquid water, the initial cold content of the snow was not a large factor in explaining the anomalous monthly rate of ablation. For perspective, the initial cold-content was less than -1 $\text{MJ}\cdot\text{m}^{-2}$ (compared to an average March value of $\approx$ -3 $\text{MJ}\cdot\text{m}^{-2}$) whereas the cumulative energy directed towards melt during March was over 100 $\text{MJ}\cdot\text{m}^{-2}$, meaning that the initial cold-content represented less than 1% of the energy directed towards ablation and is easily overcome. For some additional perspective, Jennings et al. (2018) also found that cold-content rarely exceeds -10 $\text{MJ}\cdot\text{m}^{-2}$ in cold, continental, alpine snowpacks. Both Jennings et al. (2018) and Seligman et al. (2014) also noted the timescale dependency of the relationship between cold-content and subsequent ablation.

But, the rates of overnight development of cold-content do impact the daily rate Δ SWE, since the overnight cold-content production rate must be overcome the following day before melting can begin (Fig. 5f) and is as much as one-third of the following day's rate of melt. The overnight development of cold-content is driven by LW_{net} and H_l in some limited cases, but is limited by the refreezing of meltwater overnight, which adds energy back to the snowpack (Fig. 5e).

Additional sensitivity tests that prescribed various snow initial conditions on 1 March, informed by historical snow states (Fig. S6), found that the accumulated amount of melt was largely insensitive to initial condition (or increased slightly for deeper snowpacks). Deeper snowpacks (in terms of SWE, not height) were able to retain more liquid water, meaning the change in ΔSWE (which, again, is the totality of ice and retained liquid water in the snow) during the event was mitigated by as much as TODO: 50 mm; in this way, higher antecedent SWE conditions are more resilient to snow ablation via their ability to retain more liquid water, not just initial cold-content of the snowpack. Observations show that snowpacks can retain 8—10% liquid water by volume TODO: CITE.

4.5 Future directions and connections to with climate change

In addition to the observed increases in SEHW conditions during the historical record, the frequency of heatwave events is expected to increase with climate change (Seneviratne et al., 2023). At the same time, climate models underestimate may monthly high temperature extremes by 11—-12% (Duan et al., 2025). The causes of extreme rates of snow ablation from SEHW in general, such as those reported in Rhoades et al. (2026) may occur for a variety of reasons that are not represented by the March 2026 event. In their work, a minimum of 3 days with above-freezing and +6 K above-average T_{air} was used to classify SEHW events. We find that both the late February ROS event, and the late March dry and warm period, both qualify as a SEHW following those criteria. This highlights the variety of mechanisms that may lead to anomalous snow ablation during periods of anomalous T_{air}. A complete accounting of the energy balance mechanisms during the wet-SEHW event in February is beyond the current scope, but merits investigation.

Musselman et al. (2017) noted that, given the strong dependency on SW_{net}, that snowmelt rates may decline in a future even warmer world, since snowmelt will occur earlier in the year when the sun is weaker. One might wonder if this pattern holds true for anomalous and intense heatwave events such as this one. We find that it does. The maximum rates of ablation during this event was on the order of 18 mm/day. Water year 2023, however, was an above average snowfall year and snow persisted well into July at CSSL— rates of snow ablation exceeded 70 mm/day during that time.

This work suggests that analyzing the covariance between heatwaves and sustained $SW\downarrow$ anomalies (Fig. 10) is an important feature to explore to understand SEHW impacts. Rough calculations suggest that as much as a third to a quarter of the anomalous ablation may be tied to the anomalous $SW\downarrow$ during the month of March (40 $\mathrm{W{\cdot}m^{-2}}$ higher than average). The relationship between anomalous T_{air} and $LW\downarrow$ is more complicated, however. It has previously been suggested that the relationship between $LW\downarrow$ and T_{air} may explain the efficacy of temperature index methods (Ohmura, 2001); this study does not necessarily support that conclusion, since $LW\downarrow$ anomalies are also associated with cloudy air-masses (Fig. 10). Nonetheless, examining the relationship between SEHW events and background humidity merits investigation, given the relationship with $LW\downarrow$ (Brutsaert, 1975) and sublimation.

5 Conclusions

It is "virtually certain" that climate change has already driven shifts in towards earlier heatwaves and their increased intensity and frequency (Seneviratne et al., 2023), and along with them snow-eater heatwave (SEHW) conditions (Rhoades et al., 2026).

These trends are expected to continue. However, the empirical relationships used in operational models and correlation studies between anomalously warm T_{air} and snow ablation do not have a robust physical basis, posing challenges for anticipating how snow responds to the these events, and how landscapes can be managed to provide resiliency against them.

To address this challenge, we started by investigating the anomalously warm March 2026 in the Sierra Nevada. Temperatures were three-σ above average based on high-quality records (and greater than 20 °C at the CSSL). Correspondingly, snow ablated at rapid, unprecedented rates (for 9 out of 12 SNOTEL stations) during that month. Using models, remote sensing, reanalysis, and field observations in an open and forested site, we isolated the causes of extreme ablation rate and compared them against historic conditions. Not only was March anomalously warm, the CERES satellite record (2000 to present) shows it was the sunniest March on record; $SW \downarrow$ was $+40\ \mathrm{W{\cdot}m^{-2}}$ above average, driven by a near-complete lack of low or mid-level clouds. With some assumptions, this is enough energy to melt a maximum of 100 mm of snow (or approximately 25% of the snowpack on March of that year). If there are flavors of SEWH event, this was of the "sunny and dry" variety.

Combined with the absence of new snowfall, snow α declined rapidly. The higher-than-average $SW \downarrow$ combined with well-below-average snow α meant that SW_{net} was the predominant ablation forcing in both open and forested sites. Vegetation canopy shielded the sub-canopy snow from the sun, but also enhanced $LW \downarrow$ under the canopy at an increasing rate as temperatures rose ($+3.3\ \mathrm{W{\cdot}m^{-2}}$ per °C) In this way the sub-canopy snow was more directly sensitive to warm temperatures, but less sensitive to the anomalous sunshine. Combined, models and observations agree the shading effect nonetheless prevailed, meaning that the snow ablated at a lower rate under canopy (XX mm/day) relative to the open site (XX mm/day).

Snow thermodynamics is (often) a push-and-pull between overnight cooling (cold-content production) and daytime warming; the imbalance dictates if snow ablates or persists. The cumulative cold-content production overnight was not particularly anomalous compared to past March periods in the open site—- it was the daytime energy balance that dictated the extreme rate of ablation. While the events leading up to March 2026 were anomalous, the scales involved mean that extreme monthly rate of ablation (an order of 100s of accumulated $\mathrm{MJ{\cdot}m^{-2}}$) was insensitive to reasonable variations in the initial thermodynamic state of the snowpack (which are on the order of -1 to -5 $\mathrm{MJ{\cdot}m^{-2}}$).

Sensible heating and sublimation were also relatively minor factors, and the latter contributed less than 1% of the total snow ablation. Theoretical arguments show that, over 0°C snow, sublimation is permitted only by an increasingly narrow range of atmospheric humidity values, which were not consistently maintained during the month of March. Likewise, atmospheric buoyancy effects mitigate sensible heating at high T_{air} and low winds. While nearly all land and climate models make these assumptions, more work is needed to validate turbulent exchanges into snowpacks in complex terrain and in the vicinity of vegetation canopy, where these assumptions about turbulent exchange may not hold. While snow α parameterizations performed well against observations from CSSL and remote sensing, we propose that future observations should focus on α decay processes and grain-growth mechanisms during extreme heatwave events.

Moving forward, the events of March 2026 may be a harbinger of events to come. From the perspective of peak SWE, the probability of a year like 2026 may increase from 10% under the present climate to 87% by the end-of-century, assuming business-as-usual emissions (Marshall et al., 2026). This underscoring the need to adapt water and ecosystem management to provide resiliency to such events. Synoptic analyses of snow-eater heatwave events should focus on the co-variation of heatwave conditions and the dynamics that lead to the prolonged cloud-free conditions that contributed to the anomalous radiation conditions observed during March 2026.

Open Research Section

ERA5 atmospheric data and near-surface meteorological forcings for SUMMA were downloaded from the Copernicus data store ECMWF (2023). Geopotential height anomalies and standard deviations were precomputed and provided from the supplemental material of Lee et al. (2023). Snow Property Inversion from Remote Sensing (SPIReS) near-real-time data are publicly available from the National Snow and Ice Data Center Snow Today page. The SUMMA model is publicly available on GitHub. The Suomi-NPP VNP21A1D VIIRS Version 1 Land Surface Temperature product was processed and downloaded from Google Earth Engine TODO: CITE GEE.

Acknowledgments

TODO: Enter acknowledgments here. This section is to acknowledge funding, thank colleagues, enter any secondary affiliations, and so on.

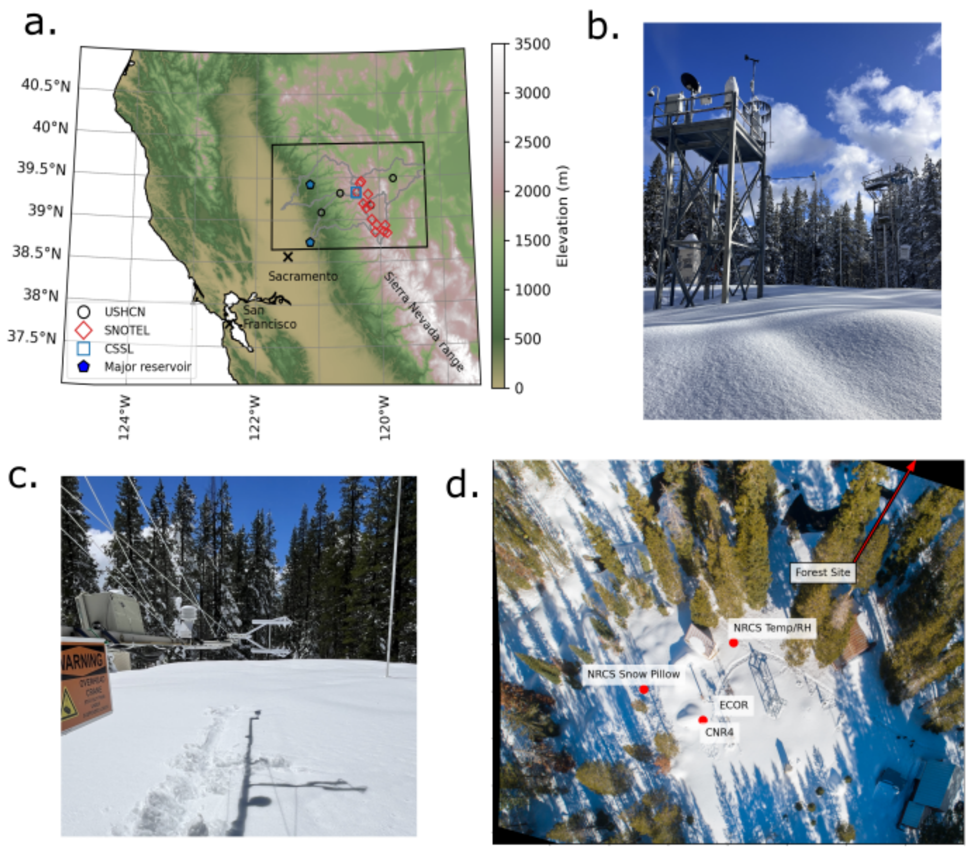

Figure 1. TODO: change the font so it's the same as the other figures(a) The northern California Sierra Nevada study region with terrain elevation. Outlines depict the Yuba, North Fork American, Lake Tahoe, and Truckee HUC8 watershed boundaries. Red open diamonds depict the location of NRCS SNOTEL sites, open black circles show the location of the four USHCRN stations in the region, and the open blue square depicts the CSSL energy balance site. (b) The CSSL meteorological tower in winter, (c) a close-up view of the Campbell CNR4 radiometer and IRGASON eddy covariance system, and (d) an aerial overview of the CSSL site. The location of the forest meteorological tower (not shown) is located just outside of the frame.

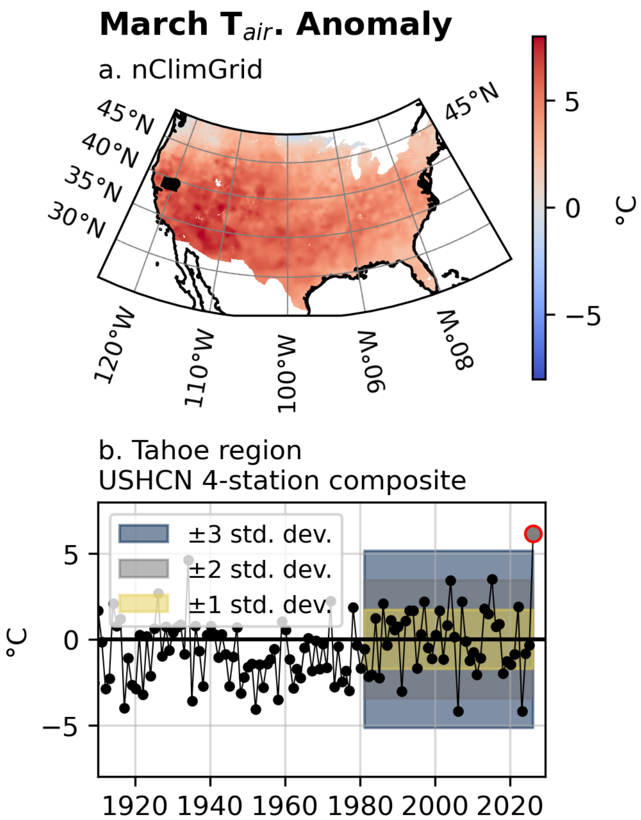

Figure 2. (a) Composite T_{air} anomalies from the four HCNv2 dataset stations in the greater Tahoe region study area for the month of March, from 1890 to 2026. Deviations from the 1980–2026 mean are depicted for one, two, and three standard deviation regions (shading). Anomalies are computed with respect to the 1980–2026 mean T_{air}. (b) The same, but for a map-view of the T_{air} anomaly of the gridded nClimGrid product.

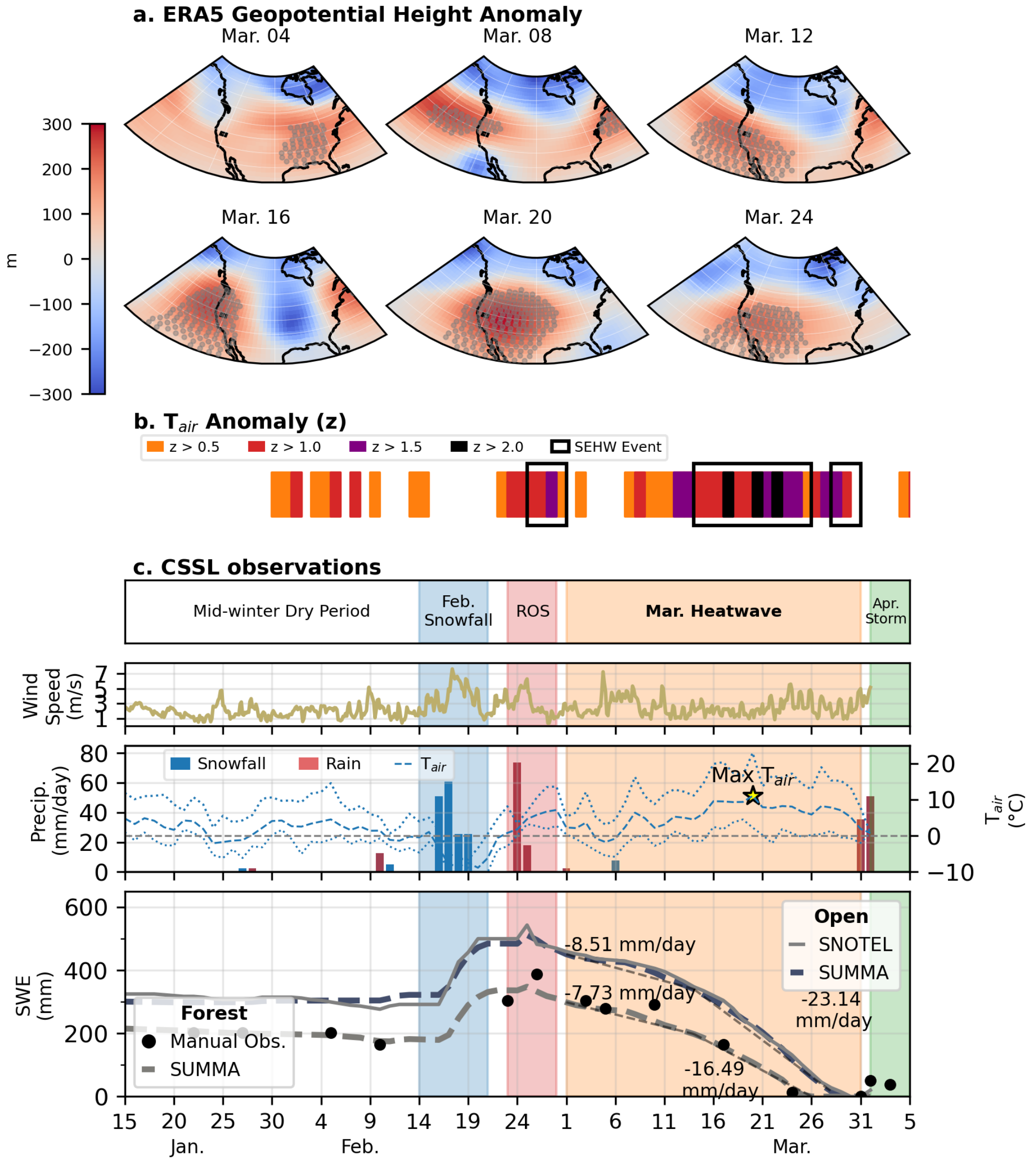

Figure 3. (a) The 500 hPa geopotential height anomaly during March 2026 heatwave from ERA5 across the WUS for six individual days. The study region is outlined using an overlain black rectangle. Gray stippling indicates regions with 500 hPa geopotential height anomalies greater than 2-σ above the TODO: 1981–2010 mean for the month of March. (b) depicts two meter T_{air} anomaly z-scores from the Tahoe City COOP station. Periods meeting the (Rhoades et al., 2026) SEHW criteria are outlined with a black box. (c) shows the time series of meteorological and snow conditions at CSSL, with different synoptic conditions highlighted by background shading (top-panel). Wind speed, precipitation with rain and snow differentiated, and two-meter T_{air} are depicted. The bottom panel depicts modeled (SUMMA) and observed SWE at the open and forest sites. Average rates of SWE loss are listed in the figure text with dashed lines superimposed on the SWE curves.

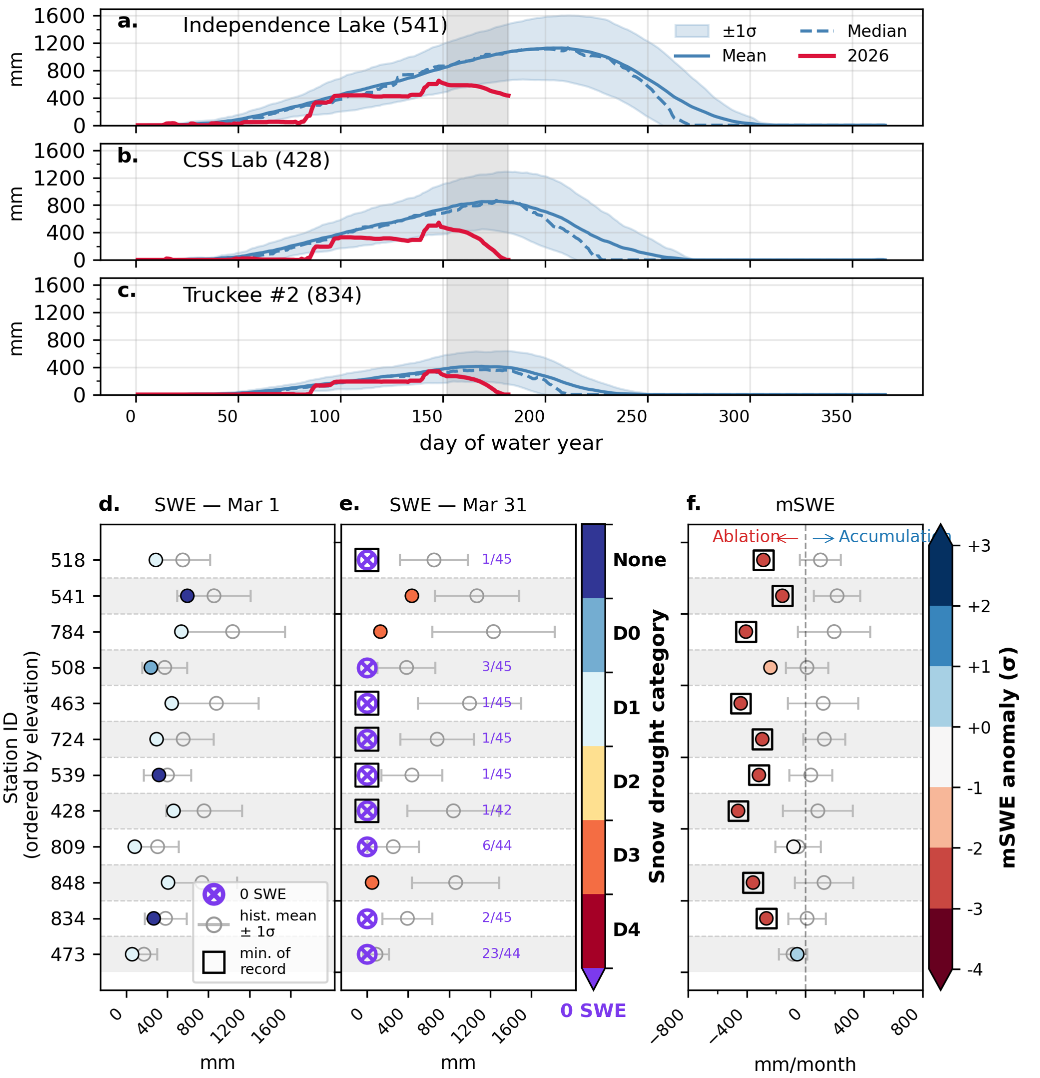

Figure 4. (a–c) A timeseries of SWE at three SNOTEL stations in the study region. The mean, median (dashed line), 1 standard-deviation (shaded region) and water year 2026 values are depicted. (d) SWE values on March 1st for the 12 SNOTEL sites in the study domain, labeled by their three-digit station code. (e) The same as (d), but for SWE on March 31. Color values correspond with the snow drought D-scale. A 0 mm of SWE value is depicted by a purple circle marker. The purple text lists the number of 0 mm SWE occurrences observed on that date relative to the period of record for that site. (f) The SWE change between Mar 31 and Mar 1 (mSWE) compared with the historical record. Color values depict the deviation from the mean in units of standard deviation. Square markers bounding symbols indicate that the SWE or mSWE is the record minimum for the period of record.

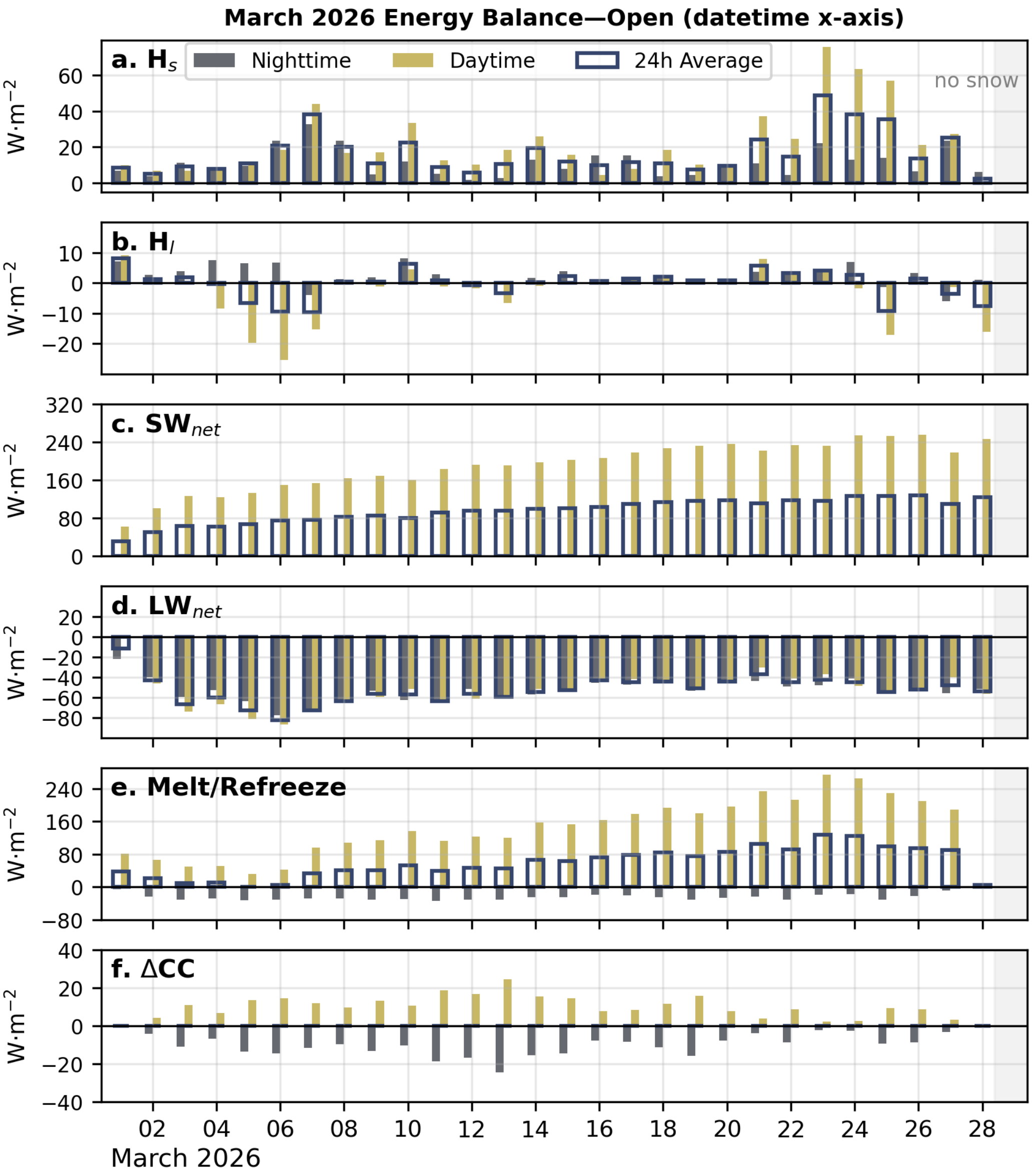

Figure 5. Daily average energy flux components including (a) H_s, (b) H_l, (c) SW_{net}, (d) LW_{net}, (e) snowpack Melt/Refreeze and (f) change in cold-content (Δ CC) during the March 2026 heatwave at the open site. Each day has three bars: the left-most yellow bar depicts the daytime flux, the right-most gray bar depicts the nighttime flux, and the outline spanning both bars depicts the 24-hour average flux. Please note that the y-scales are not equivalent across all of the figures.

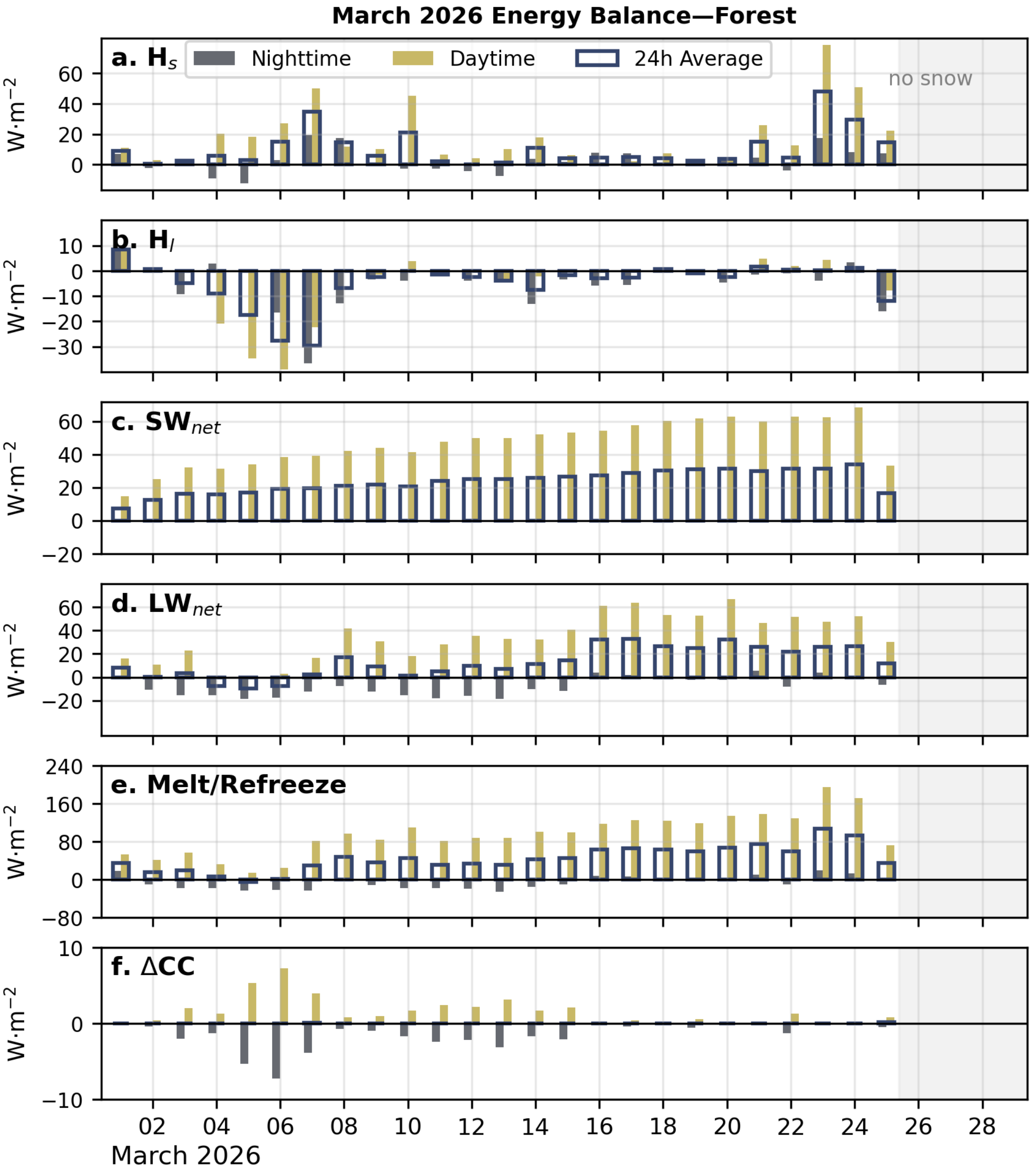

Figure 6. The same as Fig. 5 but for the forest site. Note that the y-axis limits and ticks are not the same as in that figure.

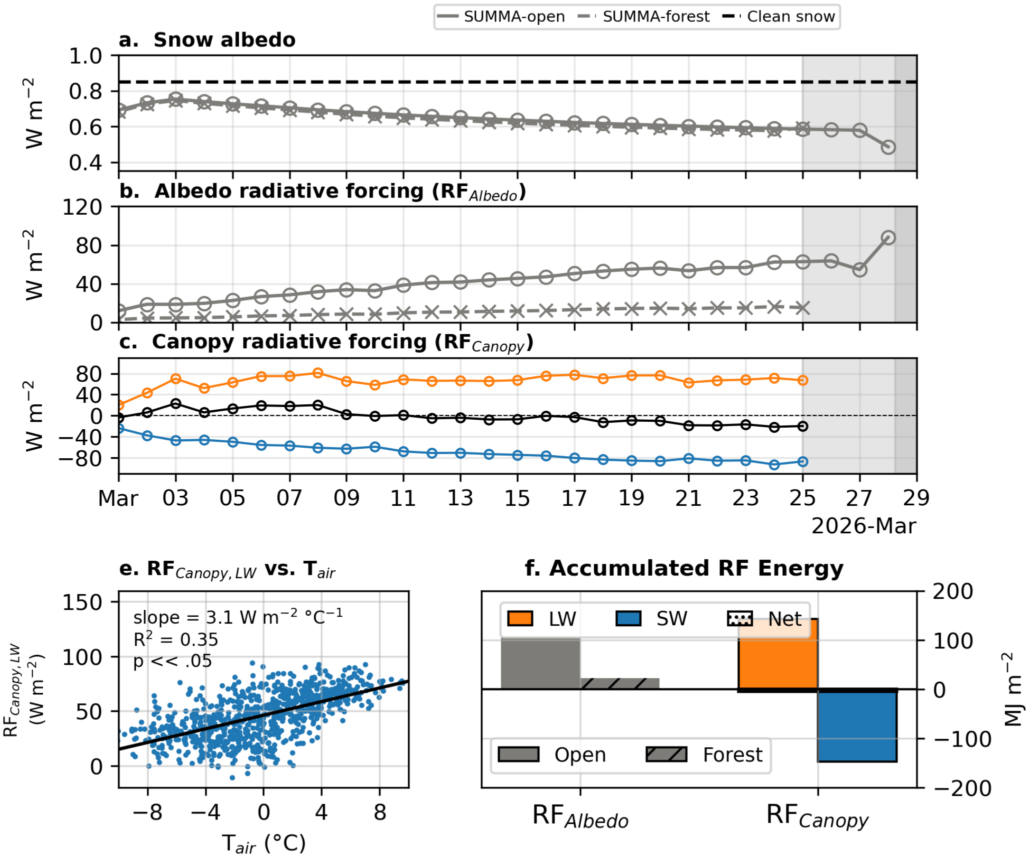

Figure 7. Timeseries of (a) α, (b) RF$_{albedo}$, (c) RF$_{canopy}$, (d) RE$_{cloud}$ for March 2026. LW and SW components are depicted in (c) and (d). Net (LW + SW) components are also depicted in (c). (e) depicts the relationship between daily-average T_{air} and the LW component of RF$_{canopy}$ for all Marches in the SUMMA historical record (2000–2026). (f) depicts accumulated energy terms during the snow-covered duration of March.

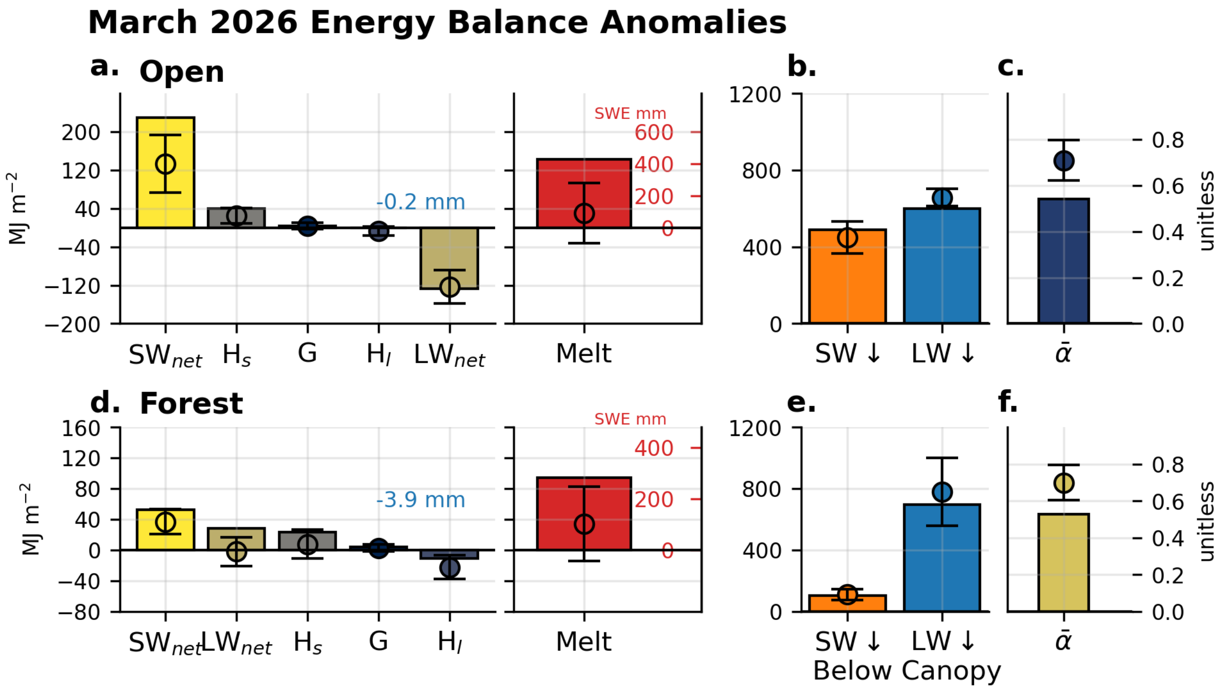

Figure 8. Accumulated snow energy fluxes (SW_{net}, LW_{net}, H_s, H_l, G), Melt, radiation forcings ($SW\downarrow$, $LW\downarrow$), and weighted snow $\bar{\alpha}$ for the open (a–c) and forest (d–f) sites for the duration of March, restricted to times when SWE is present. Open circles and whiskers show the averages and one-standard deviation from the March SUMMA historical period (2000–2026). Equivalent units of mm of melt are displayed in red text on the right-hand side of the Melt panel. The under-canopy $SW\downarrow$ and $LW\downarrow$ are depicted in panel (e).

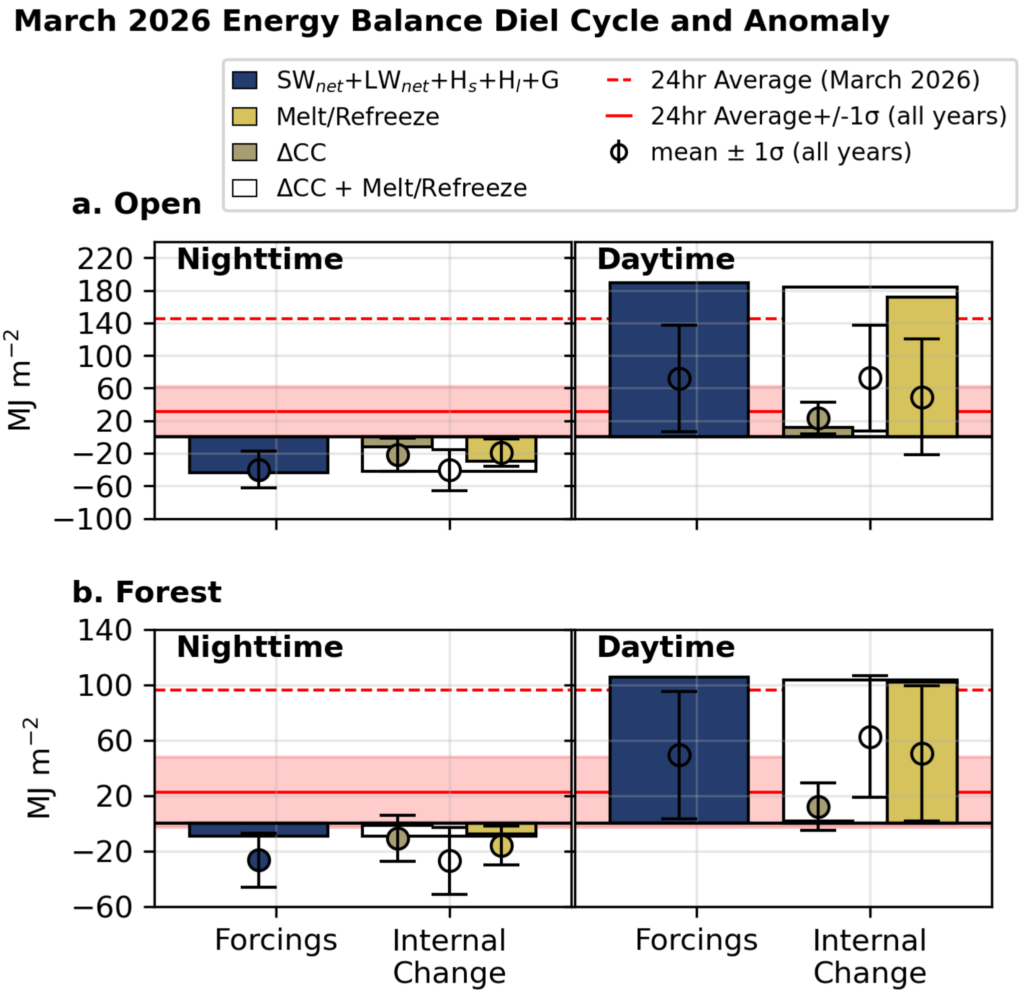

Figure 9. todo

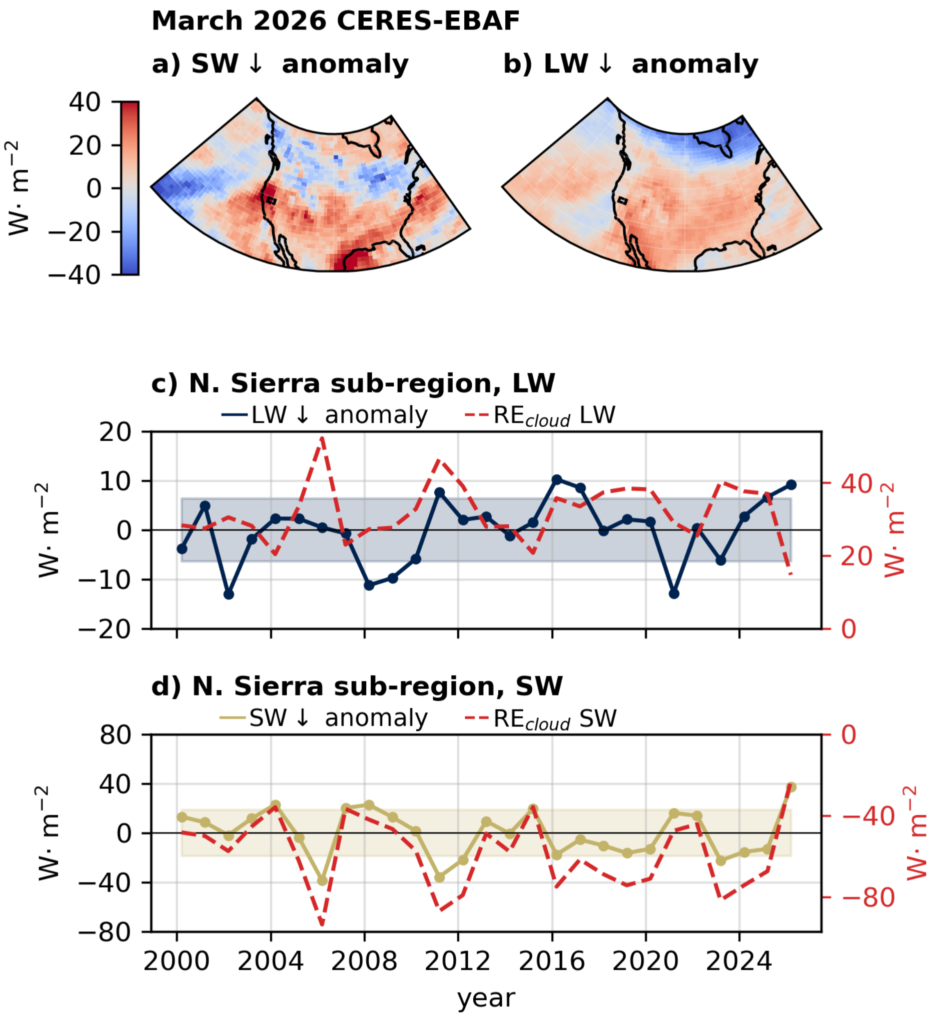

Figure 10. March $SW\downarrow$ (a) and $LW\downarrow$ (b) anomalies (relative to the March 2000 to present record) from the CERES-EBAF 1x1 degree resolution product. (c) and (d) depict the timeseries of anomalies averaged over the Sierra Nevada study region. RE$_{cloud}$ amount is depicted with a twin y-axis, with units displayed on the right-hand side in red text. The shaded region in each plot depicts one standard deviation of the anomaly.

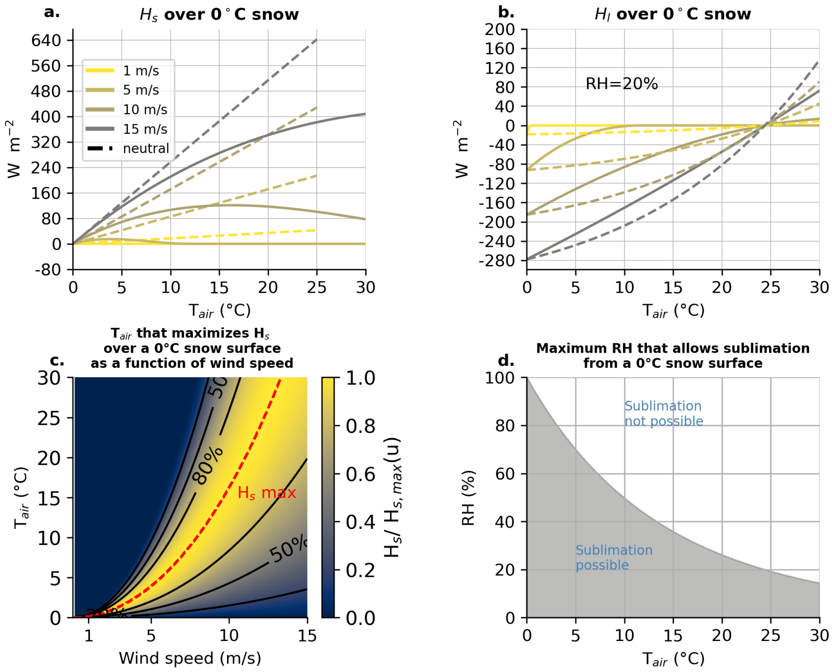

Figure 11. (a) Idealized sensible heat fluxes (H_s) into a melting snowpack as a function of $T_{\rm air}$ for four wind speeds (1, 5, 10, and 15 m s^{-1}) at a 12 m height for open-conditions and a snow surface roughness of z_0=0.002 m. Dashed lines show neutral-stability bulk transfer; solid lines show the same formulation with a stability correction applied as implemented in SUMMA. (b) the same as (a), but for H_l with a fixed RH of 20%. (c) The $T_{\rm air}$ that yields the maximum H_s into a 0 °C snowpack after correcting for stability effects (dashed red line) for a given wind speed (u). Contours depict the fraction of the maximum H_s for that wind speed value. (d) The thermodynamic limitation of sublimation as a function of $T_{\rm air}$, depicting the maximum RH that permits sublimation from a 0 °C snowpack.

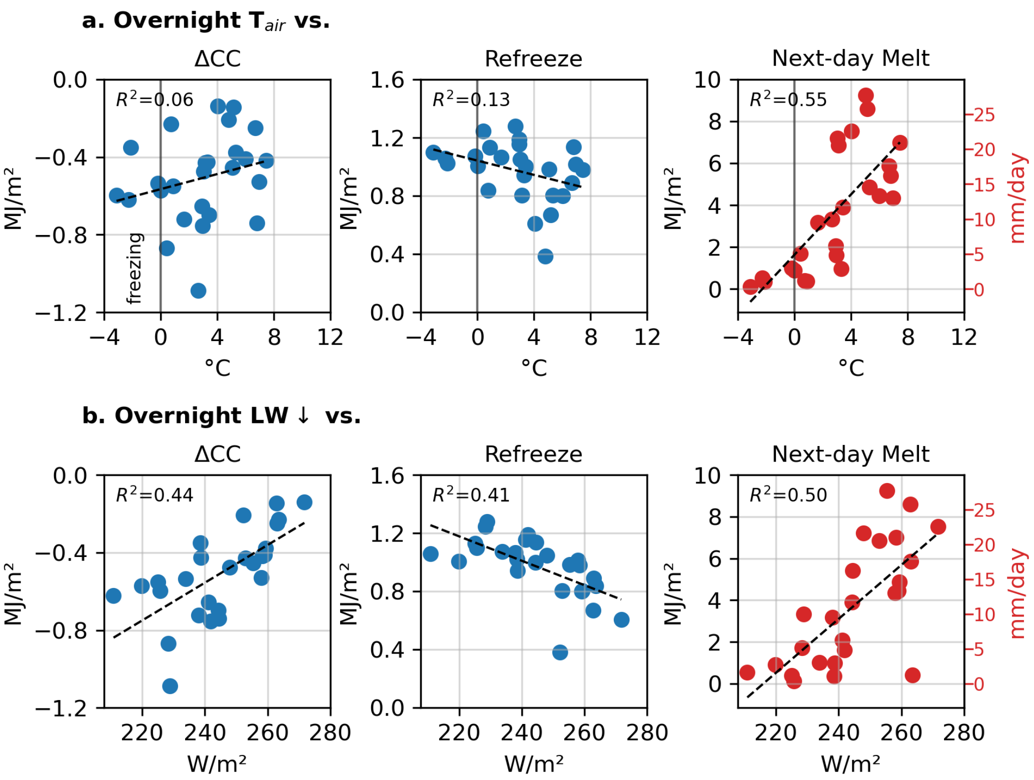

Figure 12. The predictive power of overnight average T_{air} compared against overnight average LW_{net} for ΔCC, refreezing, and total melt during the next-day. R^2 values and linear-regressions (dashed line) are depicted.

Table 2. TODO: caption: Forest and open-site energy balance terms ($MJ \cdot m^{-2}$; mm SWE equivalent in parentheses for H_l and Melt) and cold content change ($MJ \cdot m^{-2}$) for March 2026 compared against the historical climatology (mean ± std).

		SW_{net}	LW_{net}	H_s	H_l	G	Melt	ΔCC
Forest	2026	52.1	28.1	22.9	-11.0 (-3.9)	4.0	94.3 (282.4)	0.0
	Mean	37.0	-2.2	7.7	-22.5 (-7.9)	2.7	34.4 (103.1)	1.2
	Std	16.4	19.4	19.2	15.8 (5.6)	4.6	49.4 (147.8)	2.8
Open	2026	229.3	-127.5	39.7	-0.7 (-0.2)	4.3	142.1 (425.6)	0.0
	Mean	132.8	-123.3	25.0	-6.5 (-2.3)	3.6	30.0 (89.9)	2.0
	Std	61.2	35.8	16.9	9.6 (3.4)	6.3	64.1 (192.0)	5.4

Table 3. TODO: caption: Day/night decomposition of forcings (SW_{net}+LW_{net}+H_s+H_l+G), Melt/Refreeze, and cold content change (ΔCC), in $MJ \cdot m^{-2}$, for March 2026 compared against the historical climatology (mean ± std).

		Forcings		Melt/Refreeze		ΔCC	
		Day	Night	Day	Night	Day	Night
Forest	2026	105.4	-9.2	101.9	-7.5	1.7	-1.7
	Mean	49.2	-26.6	50.5	-16.0	12.2	-11.0
	Std	47.0	19.8	49.5	14.0	17.6	16.9
Open	2026	189.2	-44.1	172.1	-29.9	11.9	-11.9
	Mean	71.5	-39.9	49.2	-19.1	23.5	-21.5
	Std	66.6	23.1	72.4	16.9	19.8	20.8

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